

- [illegible]

- [illegible]

It is given that $R = \frac{4H}{\sqrt{3}}$.

- (c) Show that $\theta = 60^\circ$. [1]

[illegible]

It is given also that $u = \sqrt{40} \text{ m s}^{-1}$.

- (d) Find, by differentiating the equation of the trajectory or otherwise, the set of values of x for which the direction of motion makes an angle of less than 45° with the horizontal. [4]

[illegible]